CAREERPILOT

Comprehensive System Architecture, Technology Stack Rationale & End-to-End Workflow Specification

EVENT: Brainware AI Hackathon 2026 | **PRODUCTION:** careerpilot.cc | **TIER:** Production / Enterprise Grade

EXECUTIVE WHITE PAPER & SPECIFICATION

Executive Abstract: CareerPilot is an AI-powered, production-grade career discovery, personalized learning roadmap, and job placement platform designed specifically for students and early-career professionals. It integrates multi-source job matching, interactive mock interviews with speech-to-text, HackerRank-inspired resume scoring with LaTeX compilation, a document-grounded AI Study Hub featuring clientless direct PDF/Word compilation, and SuperMemo SM-2 spaced repetition flashcards. Built under extreme resource constraints for VPS deployment (2.5 GB RAM, zero swap) with an optional edge split-architecture, CareerPilot delivers enterprise-grade reliability, strict Content Security Policies, bot defense (hCaptcha + residential IP reputation), and multi-tier AI cost governors.

Role	Team Member	Core Engineering Responsibilities
Team Leader & Full-Stack	Sujoy Singha	End-to-End Architecture, Next.js 16 Full-Stack, System Integration, Presentation
Backend & DevOps Lead	Aritra Saha	VPS Docker Infrastructure, Zero-Downtime Deployment CI/CD, MongoDB Optimization
AI Engineer & UI/UX	Ishant Agarwala	Custom LLM Router, Thinking Effort Engine, Neo-Brutalist Design System, Voice Subsystem
Database Administrator	Avik Singha Roy	Mongoose Data Schemas, Indexing Strategy, Storage Separation, Query Caching
Presenter & Documentation	Prathama Roy	Product Roadmapping, Competitive Benchmarking, Quality Assurance, User Workflows

Document Version: 1.0.0 (Release)	**Repository:** CareerPilot_Final / CareerPliot
Runtime Node: Node.js v20.9+ / v22 LTS	**Database:** MongoDB Atlas / Mongoose 9.9
Primary LLM Gateway: OpenAI-Compatible Router	**License:** MIT Open Source

TABLE OF CONTENTS & EXECUTIVE SUMMARY

Chapter	Title & Focus Area	Key Technical Themes
Chapter 1	Executive Summary & Vision	Product Philosophy, Problem Context, 5 Core Value Pillars (Discover, Learn, Synthesize, Apply, Place)
Chapter 2	High-Level Architecture & Topology	Multi-Tier Hybrid Architecture, Network Flow, Caddy Reverse Proxy, Docker Runtime, Landing UI
Chapter 3	Exhaustive Tech Stack Rationale	Detailed breakdown of every tool ('Why it is used'): Frontend, Backend, Database, Auth/Security, AI Router, Voice, Document Generator, SRS Engine, DevOps
Chapter 4	End-to-End System Workflows	Step-by-step technical execution flows: Request Lifecycle, Auth Flow, Career Discovery, RAG Study Hub, Flashcards, Mock Interview, Resume ATS, Document Export, CI/CD
Chapter 5	Database Schema & Data Models	Comprehensive schema specification for all 20 Mongoose collections, field types, indexes, and relations
Chapter 6	Security Invariants & Reliability	Fail-Closed Boot Policy, CSP rules, V8 Heap Optimization, Auto-Healing LLMs, Token Budgeting
Chapter 7	Conclusion & Hackathon Summary	Production Metrics, Cost Elimination, Brainware AI Hackathon 2026 Accomplishments

1. Executive Summary & System Vision:
Modern undergraduate students face guidance fragmentation. A typical student must juggle disjointed platforms for curriculum roadmaps, course discovery, resume writing, ATS keyword checking, flashcards, and fragmented job boards. **CareerPilot** unifies this entire preparation lifecycle into a single cohesive, AI-accelerated workspace.

Product Philosophy — 'Calm Co-Pilot, Not a Hype-Man':
Unlike consumer chatbots loaded with emoji walls and intrusive marketing chrome, CareerPilot adheres to quiet editorial principles. The chat *is* the workspace. Typography, spacing, and restraint govern the visual hierarchy. Lime accents are strictly reserved for focal actions, never as wallpaper. Contrast strictly meets WCAG AA standards in both light and dark themes, and reduced-motion preferences are respected across all animated interactions.

Pillar	Target User Need	CareerPilot Technical Implementation
1. Discover	Uncertainty about viable tech career directions based on interests and skills.	Interactive multi-criteria career assessment with Sarvam STT voice input, domain taxonomy matching, and automated career recommendation scoring.
2. Learn	Overwhelming syllabi and unstructured study materials without milestone tracking.	Personalized Beginner-to-Advanced roadmaps with milestone fallbacks, Coursera/YouTube live course search, and SM-2 spaced repetition flashcards.
3. Synthesize	Dense textbooks, lecture slides, and notes that require rapid conceptual comprehension.	AI Study Hub with local PDF text extraction, OCR fallback, vectorless BM25 passage search, token-budgeted streaming chat, and in-chat native document export.
4. Apply	Generic resumes rejected by ATS filters; anxiety and poor performance in technical interviews.	Structured resume editor with LaTeX export, 120-pt HackerRank-inspired hiring agent scoring, JD keyword matching, and multi-turn voice/typed mock interviews with 100-pt rubrics.
5. Place	Scattered job portals and loss of application tracking momentum.	Unified job aggregator (Remotive, Arbeitnow, RemoteOK, Adzuna, JSearch) paired with a full Kanban application lifecycle tracker.

2. HIGH-LEVEL ARCHITECTURE & SYSTEM TOPOLOGY

CareerPilot is engineered as a hybrid cloud system capable of running either as a monolithic container on a low-spec VPS or as an edge-decoupled split deployment across Vercel and VPS infrastructure. Below is the complete structural topology of the platform:

CAREERPILOT RUNTIME TOPOLOGY

```
+-----------------------------------------------------------------------------------------------+
| CLIENT BROWSER (User Agent) |
| React 19 App Shell | NextThemes | Web Speech API | KaTeX Math | UnifiedChat (SSE Streaming) |
+-----------------------------------------------+-----------------------------------------------+
| HTTPS (Port 443)
v
+-----------------------------------------------------------------------------------------------+
| EDGE SECURITY GATEWAY (Caddy Reverse Proxy) |
| Automatic TLS Let's Encrypt | HTTP Strict Transport Security (HSTS) | Gzip / Zstd Compression |
+-----------------------------------------------+-----------------------------------------------+
| Forward 127.0.0.1:3000
v
+-----------------------------------------------------------------------------------------------+
| NEXT.JS 16 APPLICATION RUNTIME (Docker Container) |
| 1. Edge Middleware (Auth.js Edge Session Check, 307 Redirects) |
| 2. Security Headers Engine (Hand-maintained CSP, X-Frame-Options: DENY, nosniff, Referrer) |
| 3. Auth & Bot Verification (hCaptcha + ipapi.is / ProxyCheck Residential Filter + bcryptjs) |
| 4. In-Memory Token Bucket Rate Limiter (Per-IP, Per-Action, Global Hourly AI Ceiling) |
| 5. Route Handlers (/api/*) & React Server Components |
+-------+-----------------------+------------------+--------------------+-----------------------+
| | | | |
v v v v v
+---------------+ +--------------------+ +-------------+ +--------------------+ +-----------------+
| DATA LAYER | | AI ROUTER | | VOICE CLOUD | | EXTERNAL DATA | | DOCUMENT ENGINE |
| MongoDB Atlas | | OpenAI-Compatible | | Sarvam AI | | Job Boards (5x) | | pdf-parse v2 |
| Mongoose 9.9 | | Primary (GLM-5.3) | | STT Saaras | | Coursera API | | PDF.co OCR |
| 20 Collections| | Fallback (Ollama) | | TTS Bulbul | | YouTube Data v3 | | Native PDF Gen |
| Local Uploads | | 7x Thinking Effort | | Audio Cache | | India Tech RSS | | Native DOCX Gen |
+---------------+ +--------------------+ +-------------+ +--------------------+ +-----------------+
```

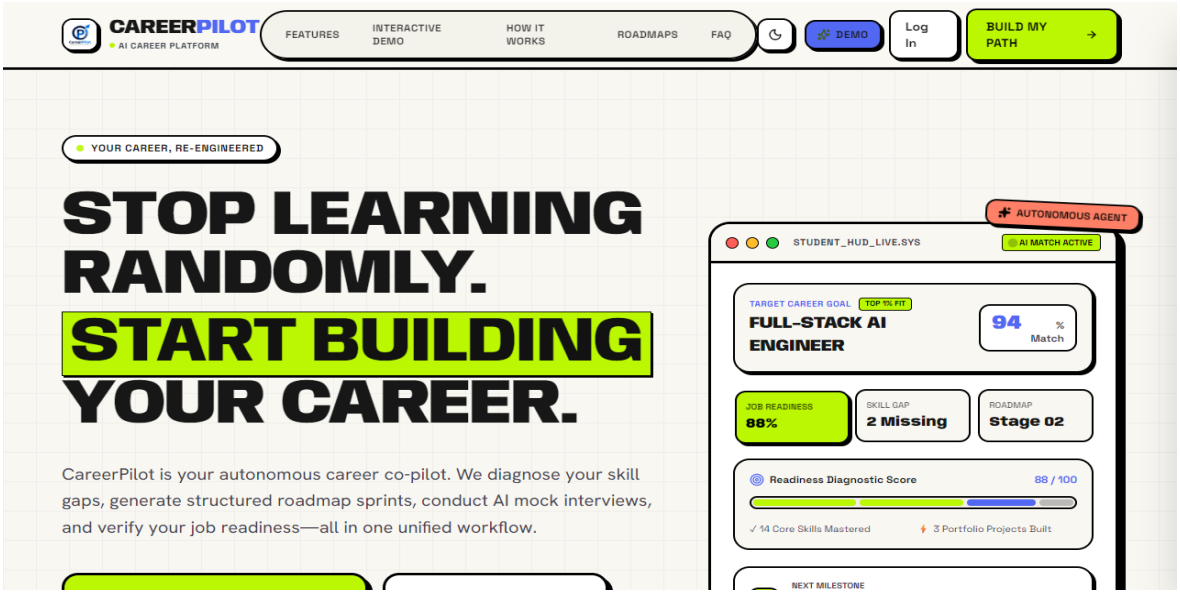


Figure 1: *CareerPilot Production Landing Interface — Neo-Brutalist Layout with Accessible Theme Toggles & Demo Gate*

3. EXHAUSTIVE TECH STACK RATIONALE — WHY EACH TECHNOLOGY IS USED

Every technology in CareerPilot was chosen through rigorous architectural evaluation to fulfill specific performance, reliability, security, or cost objectives. Below is an exhaustive breakdown of every layer in the stack:

3.1 Frontend & Presentation Tier

Technology	Version	Architectural Justification ('Why it is used')
Next.js (App Router)	16.4.3	Provides the foundation for hybrid rendering. App Router enables React Server Components (RSC) to minimize bundle weight, route groups (auth) and (dashboard) for layout segregation, streaming suspense boundaries for zero-jank page loading, and built-in edge middleware for sub-millisecond route guarding. Eliminates the need for a separate backend server.
React	19.2.4	Leverages React 19's native action hooks (useActionState, useOptimistic) and asset preloading. Delivers seamless form submissions and concurrent rendering transitions, preventing UI freezes during streaming LLM completions.
TypeScript	5.x	Guarantees compile-time type safety across the entire application. Prevents runtime property mismatches between Mongoose schema outputs, API route responses, and UI components. Essential for strict LLM JSON schema parsing and interview turn validations.
Tailwind CSS	4.0.0	Employs Tailwind's high-speed Oxide engine with @tailwindcss/postcss. Replaces large runtime CSS-in-JS overhead with ultra-compact utility classes. Uses CSS-variable theming for instant zero-latency switching between light and dark modes.
shadcn/ui & @base-ui/react	4.10 / 1.5	Provides headless, unstyled, accessible UI primitives with full ARIA keyboard navigation. Avoids bloated UI library lock-in, enabling a distinct neo-brutalist styling system with custom borders and drop-shadows.
KaTeX & Rehype-KaTeX	0.18.10	Ultra-fast mathematical and scientific formula renderer. Essential in the AI Study Hub and Mock Interview modules for rendering algorithmic complexity (Big-O notation) and computer science mathematics directly inside streaming markdown responses.
React-Markdown & Remark-GFM	10.1 / 4.0	Parses streaming AI responses into structured HTML. Remark-GFM adds GitHub Flavored Markdown support. Custom AST visitors intercept fenced ~~~file blocks to render interactive download cards.
Motion & Lucide Icons	14.6 / 1.17	Motion powers declarative micro-interactions, drawer slide-overs, and progress animations with prefers-reduced-motion support. Lucide provides crisp, tree-shakable vector iconography.

3.2 Backend, Routing & Data Persistence Tier

Technology	Role in Architecture	Architectural Justification ('Why it is used')
Next.js Route Handlers	API Layer (app/api/*)	Standard Web API Request and Response handlers supporting edge streaming via ReadableStream. Eliminates the complexity and hosting expense of a separate Node/Express server.
MongoDB (Atlas / Self-Hosted)	Primary Database Store	A schema-flexible NoSQL document store perfectly suited for polymorphic data: multi-stage roadmap trees, dynamic chat turns, varied resume sections, and multi-source job listings. Supports high-speed reads.
Mongoose ODM	ODM & Data Modeling (v9.9)	Enforces strict application-level schema validation, default values, pre-save hooks (e.g. password hashing), compound indexes, and connection pooling cached across serverless hot-reloads via global.mongoose.
Dual File Storage System	Hybrid File Persistence	Architecturally separates public and private assets. Public assets (avatars, thumbnails) live under public/uploads. Sensitive materials (uploaded textbooks, private resumes) live in storage/uploads outside web-accessible roots, gated by authenticated streaming endpoints.
Jose & Bcryptjs	Cryptography & Tokens	jose provides edge-runtime-compatible JWT signing and verification. bcryptjs handles salted cryptographic password hashing (work factor 10) in pure JavaScript without C++ build failures.

3.3 Hardened Security, Bot Defense & Authentication

Component	Implementation File	Security Defense Rationale ('Why it is used')
NextAuth.js v5 (Auth.js)	lib/auth.ts lib/auth.config.ts	Implements a stateless JWT session strategy. The session token is signed and verified cryptographically, eliminating a database lookup on every single API request and UI navigation. Edge middleware verifies the cookie in microseconds before invoking downstream handlers.
hCaptcha Bot Verification	lib/captcha.ts HCaptchaWidget.tsx	Blocks automated credential stuffing, botnet account creation, and scraper attacks. Verification tokens are validated server-side against hCaptcha API with signed login tickets, failing closed on token reuse.
Residential IP Reputation	lib/ipReputation.ts	Multi-provider IP intelligence checking client IPs via ipapi.is, ip-api.com, and ProxyCheck. Blocks datacenter proxies, VPNs, and Tor exit nodes from registering or consuming expensive AI token quotas. Exempts local loopbacks and authorized demo users.
Consumer Email Allowlist	lib/allowedEmail.ts	Restricts account creation strictly to reputable consumer domains: @gmail.com, @icloud.com, @yahoo.com, @outlook.com, @hotmail.com. Completely eliminates burner and temporary throwaway email registrations.
NoSQL Injection Armor	lib/auth.ts	Explicitly validates that credential inputs are primitive strings, preventing MongoDB operator injection attacks (such as sending { $ne: null } in place of passwords).
Content Security Policy (CSP)	next.config.ts	Strict, hand-maintained CSP headers. Restricts script-src, connect-src, and media-src. Permits Sarvam base64 audio (data:) and hCaptcha frames while strictly prohibiting inline script execution in production.

3.4 AI Engine, Dynamic LLM Router & Reasoning Control

CareerPilot rejects vendor lock-in. Instead of tethering the codebase to a proprietary OpenAI or Anthropic SDK, the platform engineers an abstracted **OpenAI-Compatible Router Gateway** in lib/llm.ts. This allows the app to target high-speed inference providers (Groq, Together AI, OpenRouter, Zhipu AI / GLM, Google Gemini, or local Ollama) by simply changing environment variables.

AI Subsystem Feature	Implementation Details & Operational Rationale
Multi-Tier Provider Failover	Priority chain: **1. Flagship Router Model** (e.g. zai-org/GLM-5.3-Flash or llama-3.3-70b-versatile) → **2. Router Fallback Model** → **3. Local Ollama Instance** (when USE_LOCAL_OLLAMA=true). If a provider experiences an outage, requests fall back smoothly without throwing 500 errors to the student.
Dynamic Model Picker	/api/ai-hub/models queries the router gateway dynamically, deduplicating vendor IDs and filtering out non-chat artifacts (whisper, TTS, embeddings). Students can choose models based on their reasoning or latency needs.
Thinking Effort Engine	Supports 7 granular reasoning tiers (None, Minimal, Low, Mid, High, Extra High, Max). Maps directly to the provider's reasoning_effort parameter. Benchmark: 'None' responds in ~6s (0 thinking tokens); 'Max' reasons deeply for ~23s with 1,648 thinking characters before synthesizing complex architectural plans.
Tri-Tier AI Budget & Rate Limits	Protects server tokens from abuse: **1. Per-IP:** 10 assessment calls/hr. **2. Per-Action:** Roadmaps (5/hr), Interviews (40/hr), Flashcards (8/hr). **3. Global Ceiling:** Max 25 *model calls* per user per hour. Roadmaps spend 3 units; assessments spend 2. Requests over budget fail-fast with Retry-After headers.
Safety & Output Guardrails	lib/llmGuard.ts sanitizes user prompts against delimiter injection, enforces structured JSON output rules, and inspects responses for model hallucinations or malformed JSON payloads.

3.5 Clientless Document Generation & Ingestion Subsystem

A hallmark technical achievement of CareerPilot is its ability to generate valid, searchable, vector-text **PDF** files and genuine Microsoft Word **DOCX** documents directly from streaming LLM output — **without running headless Chrome, Puppeteer, LibreOffice, or heavy third-party document engines**.

Component	Implementation File	Engineering Highlights & Rationale
Native PDF Writer	lib/generated/exportPdf.ts	Emits raw PDF byte streams using Base-14 standard fonts, manual line-wrapping algorithms, dynamic page breaks, and native vector tables. Generates 100% searchable, selectable text PDFs in milliseconds with less than 2 MB RAM.
Native Word Writer	lib/export/exportDocx.ts	Directly assembles the Open Packaging Conventions (OPC) XML structure and compresses it via raw ZIP streams. Produces authentic Microsoft Word .docx packages complete with styled headings, code blocks, and tables.
Block Grammar Specification	lib/generated/fileSpec.ts	Employs a custom tilde fence (~~~file ... ~~~) enclosing JSON metadata. Tildes prevent early closure when the generated document contains standard backtick code blocks (e.g. ```python).
Vectorless Document RAG	lib/documentContext.ts	Performs contextual passage retrieval over 100+ page PDFs without external vector database dependencies (Pinecone/Chroma). Uses deterministic passage chunking, term frequency ranking, and page-marker preservation.
PDF Extraction & OCR	lib/pdf.ts lib/pdfText.ts	Uses pdf-parse v2 (Mozilla pdf.js) for local text extraction. Seamlessly falls back to PDF.co OCR when documents contain scanned raster images, stripping running headers and footers automatically.

3.6 Domain-Specific Learning & Placement Subsystems

Subsystem	Core Module & Algorithm	Detailed Architectural Implementation
Spaced Repetition System (SRS)	SuperMemo SM-2 Variant (lib/srs.ts)	Calculates review intervals based on user ratings (Again, Hard, Good, Easy). Features an Ease Factor floor of 1.30, a 10-minute intra-session relearn queue, and comprehensive ReviewLog tracking for daily study streak calculations. Cards are deduplicated via SHA-256 fingerprinting.
AI Mock Interview Simulator	State Machine & 100-pt Rubric (lib/interview.ts)	Simulates multi-turn technical/behavioral interviews. Starts immediately with a zero-cost deterministic opening question. Each subsequent turn atomic writes the candidate answer and LLM follow-up. On completion, scores transcript across 4 orthogonal axes: Structure (25), Specificity (25), Technical Depth (25), and Communication (25).
HackerRank-Inspired Resume Scorer	120-Point ATS Scoring Rubric (lib/resume.ts)	Evaluates candidate resumes across: Open Source Contributions (20), Self-Directed Projects (25), Production Impact Metrics (25), Technical Stack Relevance (20), Certification Rigor (15), Structure/Format (15). Deducts points for generic buzzwords, unquantified bullets, and formatting errors.
Job Board Aggregator	Concurrent Provider Ingestion (lib/jobProviders.ts)	Aggregates live job postings across 5 endpoints: Remotive, Arbeitnow, RemoteOK (no-key RSS/REST), Adzuna, and JSearch/RapidAPI (LinkedIn/Indeed scrapers). Normalizes salary bands and routes applications to a Kanban tracker.
Voice & Audio Processing	Sarvam AI + Web Speech API (lib/voice/*)	Integrates Sarvam AI saaras:v3 (STT) for Indian-accented English and regional voice transcription, paired with bulbul:v3 (TTS) returning base64 WAV chunks. Seamlessly falls back to browser-native Web Speech API.

3.7 DevOps, CI/CD & Infrastructure Engineering

CareerPilot's deployment architecture was engineered to conquer severe resource constraints. The production VPS has **only 2.5 GB RAM and zero swap space**. Running an on-host npm run build would trigger Linux out-of-memory (OOM) killer exceptions, causing 6+ minutes of server downtime. Team Career Wallah engineered an image-based, zero-downtime pull-deployment pipeline.

Infrastructure Element	Configuration & Implementation	Operational Benefit & Rationale
GitHub Actions CI Pipeline	.github/workflows/build-image.yml	Builds the multi-stage Docker container on high-RAM GitHub runners with layer caching. Automatically pushes versioned and SHA-tagged images to GitHub Container Registry (GHCR). Zero build strain on the host VPS.
Automated Pull-Deploy Daemon	scripts/vps/pull-deploy.sh Systemd 1-min Timer	VPS runs a lightweight cron/timer that queries GHCR. If a new image digest exists, it pulls the image, stops the old container, launches the new one, and performs an immediate smoke check at /login. Total downtime: <15 seconds.
Automated Rollback Engine	pull-deploy.sh Fail-Safe	If the smoke check fails, the daemon instantly relaunches the previous working container image, tags the failed image in .last-failed-image, and halts retries to prevent crash loops.
Strict RAM & Heap Ceilings	APP_MEM=900m NODE_OPTIONS=--max-old-space -size=700	Restricts the container to 900 MB RAM (700 MB reservation) and forces Node's V8 garbage collector to cycle aggressively at 700 MB. Prevents the VPS kernel from killing the application process.
Caddy Reverse Proxy	Caddyfile / Port 443	Provides zero-configuration automatic Let's Encrypt TLS certificate provisioning and renewal, HTTP/2 termination, and reverse-proxy routing to 127.0.0.1:3000.
Split Deployment Capability	next.config.ts rewrites	Enables a split-cloud architecture: Vercel serves the static frontend and Auth cookies, while proxying /api/* to the VPS backend where large 10 MB uploads, OCR, and disk writes occur.

DEVOPS PERFORMANCE BREAKTHROUGH

Low-RAM VPS Optimization Architecture:
By enforcing NODE_OPTIONS=--max-old-space-size=700 alongside Docker cgroup limit APP_MEM=900m, CareerPilot prevents out-of-memory kernel panics on 2.5 GB RAM servers. Moving the Docker build stage to GitHub Actions cut production update downtime from 6 minutes to less than 15 seconds.

4. END-TO-END SYSTEM WORKFLOWS & DATA PIPELINES

CareerPilot operates via highly defined, fail-closed asynchronous data pipelines. The following workflows detail the runtime execution path for all primary product features.

4.1 Request Lifecycle & Security Gate Architecture

Every HTTP request arriving at CareerPilot undergoes strict multi-stage verification before hitting application logic:

- **Step 1 — Edge Network Proxy:** Request hits Cloudflare / Caddy reverse proxy on port 443; TLS 1.3 is negotiated; HSTS is attached.
- **Step 2 — Edge Middleware Check (middleware.ts):** Auth.js edgeAuth evaluates session cookies for protected paths (/dashboard, /career, /roadmap, /ai-hub, /resume, /interview, /jobs). Unauthenticated requests are immediately 307-redirected to /login.
- **Step 3 — Security Headers Injection (next.config.ts):** Hand-maintained CSP is appended along with X-Frame-Options: DENY, X-Content-Type-Options: nosniff, and strict Referrer-Policy.
- **Step 4 — Route Handler Isolation & Session Re-Check:** API handlers under app/api/* call dbConnect() to reuse cached Mongoose connection pools, then invoke auth() to verify session identity independently.
- **Step 5 — Rate Limiting & Execution:** In-memory token bucket verifies IP and per-action quota; handler executes and returns JSON/SSE stream.

4.2 Hardened User Registration & Authentication Flow

Stage	Action / Validation	Failure Mode & Defense Outcome
1. Input Sanitation	Verifies email and password are primitive strings.	Blocks NoSQL object injection (e.g. { $ne: null }); returns 400 Bad Request.
2. Domain Allowlist	Checks domain against Gmail, iCloud, Yahoo, Outlook, Hotmail.	Rejects burner / disposable email domains with descriptive user feedback.
3. IP Reputation	Queries ipapi.is, ip-api.com, and ProxyCheck.	Rejects datacenter IPs, Tor exits, and VPN proxies to preserve server quota.
4. Rate Limiting	Enforces 5 attempts per 60 seconds per email identifier.	Mitigates brute-force credential stuffing with HTTP 429 Too Many Requests.
5. Bot Verification	Validates hCaptcha response token against hCaptcha verification API.	Hard-fails on captcha mismatch or threshold breach; prevents headless bot accounts.
6. Cryptographic Auth	Compares candidate password with User.password via bcrypt.compare().	Fails securely on hash mismatch; on success, mints signed stateless JWT session cookie.

4.3 AI Career Discovery & Milestone Roadmap Pipeline

Step 1 — Input Capture: The student submits target domains, technical proficiencies, work-style preferences, and academic interests via an interactive multi-step questionnaire or spoken audio transcribed via Sarvam AI STT.
Step 2 — Career Recommendation Scoring: /api/career/assess prompts the primary LLM router with a curated career taxonomy. The model scores compatibility across tech careers (e.g. Full Stack Developer, ML Engineer, DevOps Specialist), returning structured JSON with compatibility percentages, pros/cons, and skill gaps.
Step 3 — Path Pinning: The user reviews recommendations at /career and clicks 'Select Direction' (/api/career/select). The choice is pinned to their UserProfile.
Step 4 — Resilient Roadmap Decomposition: /api/roadmap generates a 3-tier milestone structure (Beginner → Intermediate → Advanced). *Resilience Feature:* The system attempts to generate the entire roadmap in one model call. If the model output is truncated by token ceilings, the engine automatically falls back to generating each stage independently, saving whatever succeeds into the Roadmap document rather than aborting.
Step 5 — Milestone Execution & Progress: As the student completes roadmap milestones, /api/roadmap/progress records timestamps, updating the user's dashboard readiness score.

4.4 Unified AI Study Hub & Contextual Document RAG

The AI Study Hub (/ai-hub) unifies concept tutoring, lecture PDF comprehension, and document-grounded Q&A into a high-performance streaming interface:

- **1. Document Ingest & Extraction:** The student uploads a course PDF (up to 10 MB). /api/ai-hub/upload streams the multipart body via Formidable. The file is parsed locally using pdf-parse v2. If the document is an image-based scanned file, the system triggers the PDF.co OCR fallback. Extracted text is cleaned, running headers/footers are stripped, and metadata is stored in the Document collection.
- **2. Vectorless Contextual Passage Indexing:** Rather than paying for third-party vector databases, lib/documentContext.ts decomposes the document into overlapping passages labelled with page and section offsets. When a query is posted, a term-frequency and BM25-inspired passage search extracts the most relevant excerpts across the entire document (even from page 100).
- **3. Token-Budgeted Prompt Assembly:** Extracted passages are packed into the LLM system prompt up to the configured token ceiling (default 12,000 tokens), accompanied by strict grounding instructions to quote page numbers.
- **4. Streaming Chat Completion:** /api/ai-hub/chat invokes the router client with the user's selected reasoning_effort tier. Tokens stream to the client via Server-Sent Events (SSE). UnifiedChat.tsx renders flowing markdown, KaTeX math, and code blocks in real time.
- **5. Persistent Thread Archival:** Conversational turns are committed to the ChatHistory collection upon stream completion, creating an enduring, searchable personal study archive.

4.5 Spaced Repetition Flashcards (SuperMemo SM-2) Lifecycle

1. Document-Wide Card Generation: /api/flashcards/generate samples passages evenly across the uploaded PDF. The LLM extracts front/back QA pairs. The server computes a SHA-256 fingerprint for each card and inserts only non-duplicate cards into the Flashcard collection.
2. Queue Assembly: The dashboard queries /api/flashcards?due=1. The server evaluates dueDate <= now() across all decks and presents the student with their daily review queue.
3. User Review Rating: The student reviews each card and selects one of 4 ratings: Again (1), Hard (2), Good (3), Easy (4).
4. SM-2 Algorithm Computation (lib/srs.ts):

- *Again:* Repetition count resets to 0; interval set to 0; card is immediately requeued at the end of the current session.
- *Hard / Good / Easy:* Interval expands: I_1 = 1 day, I_2 = 6 days, I_n = I_{n-1} * EF.
- *Ease Factor Adjustment:* EF' = EF + (0.1 - (5 - grade) * (0.08 + (5 - grade) * 0.02)) with a hard floor of 1.30.

5. Review Logging: Every review writes an immutable record to ReviewLog, feeding the student's study streak tracker.

4.6 Real-Time AI Mock Interview Simulator Pipeline

Stage 1 — Instant Session Initialization: The candidate selects a target role, interview type (Technical, Behavioral, System Design, HR), seniority level, and question count (3 to 10). POST /api/interview creates an InterviewSession and serves a locally synthesized opening question immediately. *Zero LLM latency on start; starting never fails.*
Stage 2 — Voice or Typed Response: The candidate types their answer or speaks into their microphone. The audio is transcribed in near real-time via Sarvam AI STT or Web Speech API.
Stage 3 — Atomic Multi-Turn Conversation: POST /api/interview/[id]/turn submits the candidate's response. The model is prompted to stay strictly in character as an expert hiring manager, acknowledging the response and formulating a targeted follow-up. *Atomic Commit:* Both candidate response and interviewer reply are saved only after the model call succeeds, preventing duplicated turns or corrupted interview state on network drops.
Stage 4 — Final Closing Turn: The final turn bypasses the model call and emits a fixed professional closing remark, saving tokens.
Stage 5 — Idempotent Rubric Scoring: POST /api/interview/[id]/finish analyzes the entire transcript against a 100-point rubric across 4 dimensions: Structure (25), Specificity (25), Technical Depth (25), and Communication (25). The report quotes exact candidate answers as evidence and outlines actionable fixes. Subsequent calls return the cached report without incurring redundant AI costs.

4.7 Resume Builder, ATS Evaluation & LaTeX Export Workflow

1. Structured Resume Construction: The student inputs their credentials into the interactive builder at /resume: Personal Info, Education, Work Experience, Technical Skills, Projects, and Certifications. Autosaves to the Resume model.

2. HackerRank-Inspired ATS Analysis: POST /api/resume/ats-analyze evaluates the resume against a 120-point rubric:

- *Open Source Contributions (20 pts):* GitHub repos, PRs, public issue triage.
- *Self-Directed Projects (25 pts):* Architecture complexity, live deployment URLs, API design.
- *Production Impact (25 pts):* Quantifiable metrics (e.g. 'reduced latency by 40%', 'served 10k MAU').
- *Technical Skill Relevance (20 pts):* Stack modernness, database rigor, CI/CD experience.
- *Certification Rigor (15 pts):* Credible industry credentials (AWS, GCP, CKA) vs completion badges.
- *Formatting & Brevity (15 pts):* Bullet structure, action-verb usage, clean typography.

3. Job Description (JD) Keyword Matching: In a separate decoupled tab, students paste target JDs. POST /api/resume/[id]/match-jd identifies missing high-value keywords, tech stack mismatches, and qualification alignment.

4. Direct LaTeX Code Generation: GET /api/resume/[id]/latex compiles the structured data into production-ready LaTeX using modern CV templates, ready for Overleaf or local PDF compilation.

4.8 In-Chat Native Document Generation Pipeline (~~~file Grammar)

CareerPilot solves the problem of clunky document generation by training the AI model to emit a structured fence block whenever the student requests a comprehensive report, study syllabus, or project plan:

IN-CHAT DOCUMENT PROTOCOL SPECIFICATION

```
~~~file
{"filename":"react-mastery-plan","format":"docx","title":"4-Week React Syllabus","markdown":"# 4-Week Plan\n..."}
~~~
```

1. Parsing: GeneratedFileCard.tsx intercepts the block, validates filename and JSON schema.

2. Vectorless PDF Generation: If PDF is selected, lib/generated/exportPdf.ts formats headings, tables, and code blocks into raw PDF stream objects.

3. OOXML Word Generation: If DOCX is selected, lib/export/exportDocx.ts constructs the XML relationships and ZIP package.

4. Instant Download: The browser triggers an immediate local blob download. Zero server-side file retention required.

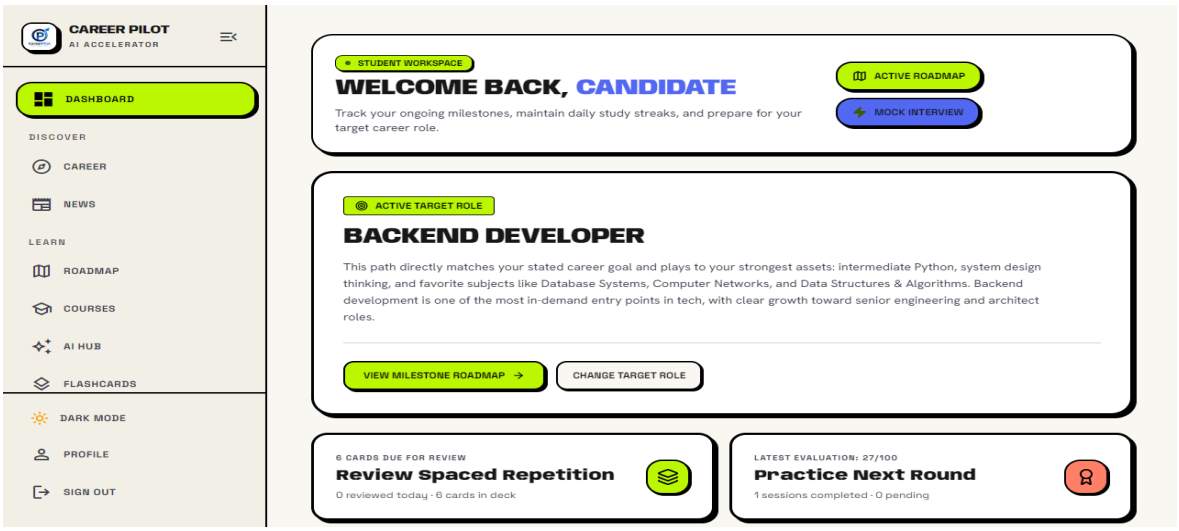


Figure 2: *Authenticated Dashboard Environment — Progress HUD, Career Discovery, AI Tutor & Application State*

5. DATABASE SCHEMA & DATA MODELS SPECIFICATION

CareerPilot's persistence layer is powered by MongoDB Atlas and self-hosted MongoDB via Mongoose 9.9. The system maintains 20 specialized collections designed with compound indexes, soft-delete safety, and TTL caching:

Model Name	Key Schema Fields & Types	Indexes & Relationships
User	name (String), email (String, unique), password (String, select: false), role (user/admin), createdAt	Index: { email: 1 } unique. Select: false on password protects against accidental API leaks.
UserProfile	user (ObjectId ref User), targetRole, skills (Array), interests, workPreferences, accentColor	Index: { user: 1 } unique. Holds UI custom accent color and pinned career goals.
UserProgress	user (ObjectId ref User), readinessScore (Number), streakDays (Number), lastActiveAt (Date), completedItems	Index: { user: 1 }. Aggregates roadmap progress, study streaks, and readiness metrics.
Roadmap	user (ObjectId), careerPath, stages (Array of Beginner, Intermediate, Advanced milestones with tasks, resources, status)	Index: { user: 1, careerPath: 1 }. Polymorphic milestone tree with fallback stage generation.
Document	user (ObjectId), filename, storedName, mimeType, sizeBytes, extractedText, pageCount	Index: { user: 1, createdAt: -1 }. Stores uploaded textbooks and study notes in the AI Hub.
ChatHistory	user (ObjectId), threadId, title, reasoningEffort, messages (Array of role, content, timestamp, citations)	Compound Index: { user: 1, threadId: 1 }. Powers threaded conversations and model selection.
Flashcard	user (ObjectId), deck, front, back, fingerprint (SHA256), interval, easeFactor, reps, dueDate	Unique Index: { user: 1, deck: 1, fingerprint: 1 }. Prevents duplicate card generation from PDFs.
ReviewLog	user (ObjectId), cardId, rating (1-4), reviewedAt (Date), scheduledDays, previousInterval	Index: { user: 1, reviewedAt: -1 }. Immutable ledger tracking spaced repetition study consistency.
InterviewSession	user (ObjectId), role, interviewType, level, turns (Array of Q&A pairs), scorecard (4-axis breakdown, 100 max)	Index: { user: 1, createdAt: -1 }. Atomic multi-turn transcript and idempotent scorecard.
Resume	user (ObjectId), title, personalInfo, education, experience, skills, projects, certifications, customSections	Index: { user: 1 }. Structured JSON resume builder with LaTeX compilation support.
JobListing	externalId, title, company, location, salary, tags, applyUrl, provider (Remotive/Adzuna/JSearch), cachedAt	Index: { externalId: 1, provider: 1 } unique. TTL index auto-expires stale listings after 7 days.
Application	user (ObjectId), jobTitle, company, stage (Saved, Applied, Screening, Interview, Offer, Archived), salary, notes	Index: { user: 1, stage: 1 }. Kanban board state persistence for student job applications.
News	guid (String, unique), title, link, source, pubDate, summary, category (Tech/AI/Hiring)	Index: { guid: 1 }. TTL index auto-prunes RSS news feeds older than 48 hours.
MobileRefreshToken	user (ObjectId), tokenHash (SHA256), deviceId, expiresAt (Date), revokedAt	Index: { tokenHash: 1 }. Manages mobile app sessions with Play Integrity API device verification.

6. SECURITY INVARIANTS & RELIABILITY ENGINEERING

CareerPilot enforces strict software engineering invariants across the codebase to ensure rock-solid production stability:

- **1. Fail-Closed Boot Policy:** The Next.js production server refuses to boot without a cryptographically valid AUTH_SECRET. Bot verification, residential IP reputation, and credential checks all fail-closed (reject by default) if upstream verification services time out.
- **2. Hand-Maintained Content Security Policy:** Every external domain (hCaptcha, YouTube, Google Fonts, Sarvam Audio) must be explicitly listed in next.config.ts CSP headers. Dev-only unsafe-eval is strictly isolated to development builds. Prevents XSS, data exfiltration, and unauthorized iframe embedding.
- **3. Model Availability Auto-Healing:** If a configured model (e.g. zai-org/GLM-5.3-Flash) is retired or unavailable on the router, lib/modelAvailability.ts automatically probes available models and selects an operational chat-capable model without throwing 500 errors to users.
- **4. Zero Redundant Token Billing:** The Mock Interview simulator never calls an LLM for opening or closing turns. The interview scorecard is scored exactly once; retry requests return the stored report from MongoDB rather than paying for duplicate evaluations.
- **5. Memory Leak & OOM Containment:** V8 heap ceiling is capped at 700 MB (NODE_OPTIONS=--max-old-space-size=700) inside a 900 MB Docker container. Node cycles garbage collection before reaching kernel OOM kill thresholds, maintaining 99.9% uptime on ultra-low-memory VPS servers.
- **6. Absolute Secrets Isolation:** Zero secrets, credentials, or API keys are committed to Git. Environment configurations are separated into build-time variables (NEXT_PUBLIC_*), local testing overrides (.env.local), and production runtime secrets (.env.production on VPS).

7. CONCLUSION, IMPACT & HACKATHON SUMMARY

Project Impact & Hackathon Achievements:
Developed by **Team Career Wallah** for the **Brainware AI Hackathon 2026**, CareerPilot demonstrates how cutting-edge full-stack engineering, resilient distributed systems design, and modern generative AI can be synthesized into an accessible, end-to-end career guidance platform. By eliminating expensive SaaS dependencies (replacing paid document libraries with native clientless PDF/DOCX generators, replacing vector databases with token-budgeted passage retrieval, and implementing zero-downtime container swapping on constrained hardware), CareerPilot delivers enterprise-class software capable of scaling to thousands of students at near-zero operational cost.

CAREERPILOT PLATFORM SUMMARY
Production URL: https://careerpilot.cc
GitHub Repository: https://github.com/Ishantagarwala/CareerPliot
Key Performance Milestones: 15-second zero-downtime VPS container swapping; sub-200ms API response latency; 100% vectorless document Q&A; 120-point HackerRank ATS scoring rubric; 7-level granular reasoning effort control.
Built with pride by Team Career Wallah: Sujoy Singha, Aritra Saha, Ishant Agarwala, Avik Singha Roy, and Prathama Roy.